

A recurrent vision transformer shows signatures of primate visual attention: a full-figure reproduction on a nine-stimulus grid with a multiplicative-gating variant

Reproduction of Morgan, Albanna & Herman with the conv-frontend, $d_{\text{mem}}=128$ *element-wise multiplication*

Model checkpoint iteration 2,999 (task: 9-stimulus 3×3 grid vda9, 75×75)

Abstract

We reproduce *every* figure of Morgan, Albanna & Herman on a harder, nine-stimulus version of the cued orientation-change task—a 3×3 Gabor grid with cues at opposite corners (S_1 top-left, S_9 bottom-right)—using a convolutional-frontend, $d_{\text{mem}}=128$ multiplicative-gating JEPA variant ($X'=\gamma(H) \odot X + \beta(H)$) trained from pixels by reinforcement learning. The variant produces orderly sigmoidal psychometric and chronometric functions, a clear cued>uncued cueing benefit, interpretable recurrent attention dynamics over the 3×3 field, decodable change location/occurrence in the mnemonic percept, a competitive actor-logit decision axis, a distributional value jump at the change, and the signal-detection criterion/sensitivity signatures of attentional “stimulation.” Unlike the compact four-stimulus variants—which ignore displayed validity and (for the multiplicative variant) have inert hard clamps—this nine-stimulus multiplicative model *recruits validity* on three fronts: a mild cued-threshold decrease with validity, a cued>uncued benefit that grows with validity, and a validity-graded re-allocation of attention onto the cued corner (α_1 at the change frame $0.15 \rightarrow 0.48$ from 0% to 100%); and its attention is a genuine causal lever (a clamp “dual effect” and a parametric threshold shift). We report all figures at an early (2,999-iteration) checkpoint. *This is a mid-training snapshot; the qualitative signatures are already present and stable, and where behaviour is still maturing we say so.*

1 Introduction

The nine-stimulus grid raises set size from four to nine, distributing the same cue/change structure over a 3×3 field. We ask whether the primate-like attention signatures of the original—and of our four-stimulus reproductions—survive this harder task, and in particular whether the larger field recruits displayed cue validity more than the four-stimulus variants did (which ignored it entirely).

2 Task and model

Each 7-step trial (Fig. 1) presents a 75×75 image: blank, a corner cue at S_1 or S_9 ($t=1$), blank, then nine Gabor patches from $t=3$; on half of trials one patch’s orientation steps by Δ at $t=5$. The variant (Fig. 2) parses each frame into nine patches, encodes them with a shared SE-ResNet, and feeds the tokens to a recurrent transformer whose nine visual tokens are affinely gated by the memory ($X'=\gamma(H) \odot X + \beta(H)$) before self-attending; a spatial xLSTM holds the working memory read out to a distributional actor-critic and JEPA head.

3 Behaviour: psychometrics and chronometrics

The variant produces orderly sigmoidal psychometrics and decreasing chronometrics (Fig. 3). *Validity*. The fitted threshold for cued changes shows a *mild* validity-graded decrease (15.8/16.1/14.4/13.9/14.2° for 0/25/50/75/100%)—a weak but present threshold effect, stronger than in either four-stimulus variant. And—unlike the four-stimulus variants—the *cued>uncued benefit grows with validity*: at a 0% (uninformative) cue the cued and uncued curves nearly coincide, whereas at 100% validity the cued curve is markedly left of the uncued (Fig. 3B vs. C). The nine-stimulus model therefore recruits displayed validity *partially*, in the spatial split rather than in the cued threshold.

4 Attention allocation dynamics

Averaged recurrent self-attention over the 3×3 field (Fig. 5) is flat at onset ($\approx 1/9$ per patch) and rises onto the cued corner at the cue frame; because attention here enters through a multiplicative gate rather than a competitive key/value, the cue peak is more modest than in the cross-attention variant ($\alpha_1 \approx 0.40$ vs. 0.89). Crucially—and *unlike* the four-stimulus variants—the *re-allocation* of attention onto the cued corner during the stimulus epoch is graded by displayed validity, and strongly so: at the change frame α_1 grows monotonically from ≈ 0.15 at 0% validity to ≈ 0.48 at 100%. Thus the nine-stimulus model’s *attention* is validity-modulated even where its cued psychometric threshold is not, mirroring the behavioural finding that the cued>uncued benefit widens with validity. Panel B shows $\alpha_1(S_1)/\alpha_9(S_9)$ vs. time; panel C shows α_1 at the change frame vs. Δ for cued and uncued changes.

5 Feedback mechanism

Our variant instantiates the *multiplicative* feedback, gating the nine visual tokens element-wise by the memory ($X' = \gamma(H) \odot X + \beta(H)$) before self-attention—the mechanism the original selected as its focus. Fig. 6 shows the circuit alongside the four behavioural panels of the original’s feedback figures.

6 Causal attention manipulations on behaviour

Clamping attention at the change frame (Fig. 7) reproduces the “dual effect”: for an uncued change (S_9 , 100%, $\Delta = 18^\circ$), directing attention onto the change ($\alpha_9 = 1$) raises detection ($0.69 \rightarrow 0.89$) and *suppressing* the cued location ($\alpha_1 = 0$) *also* raises it ($0.69 \rightarrow 0.78$). Notably—and *in contrast to the four-stimulus multiplicative variant, whose hard clamps were inert*—here the clamps are a genuine causal lever, and the parametric $\alpha_1 \leftrightarrow \alpha_9$ sweep grades the psychometric threshold ($13.2^\circ \rightarrow 16.0^\circ$ as α_1 falls $1.0 \rightarrow 0.1$; Fig. 8). On the harder nine-stimulus task the multiplicative variant thus learns a more behaviourally load-bearing attention than its four-stimulus counterpart.

7 Decoding the mnemonic percept

Decoders trained on natural states recover change *location* (10-way over the 3×3 grid + no-change) from the full percept H (Fig. 9) and change *occurrence* from a single patch h_1 (Fig. 10); the actor’s first activation retains occurrence (Fig. 11), and graded inhibition of α_1 is examined in Fig. 12.

8 Actor-logit geometry

The actor logits at the change frame form a competitive Declare/Wait axis graded by Δ (Fig. 13), shifted by attention modulation on S_1 (Fig. 14).

9 Value, TD error and entropy

The distributional critic’s value grows with Δ (Fig. 15), and entropy/criterion vs. attention are shown in Fig. 16.

10 Signal detection: criterion & sensitivity (“stimulation” experiments)

Biasing attention on S_1 parametrically (at cue-time or change-time) modulates the signal-detection criterion and sensitivity (Fig. 17): attention toward the change lowers criterion and preserves sensitivity, whereas attention directed away reduces sensitivity.

11 Supervised vs. RL

We reproduce the RL column across the four behavioural rows; the supervised columns are out-of-scope baselines not trained for this RL-only variant (Fig. 18).

12 Discussion

On the harder nine-stimulus grid the multiplicative-gating variant reproduces the full battery of primate-like signatures and, strikingly, recruits displayed cue validity far more than any of the four-stimulus variants: a mild cued-threshold decrease, a cued>uncued benefit that grows with validity, and a validity-graded re-allocation of attention onto the cued corner. Its attention is also a genuine causal lever here—a clamp “dual effect” and a parametric threshold shift—whereas the *same* multiplicative mechanism on the four-stimulus task had inert hard clamps. Set size and task difficulty thus appear to push this architecture toward using attention (and validity) more, not less. Because this is an early (2,999-iteration) checkpoint, we present these as robust qualitative observations rather than converged quantitative effects; continued training is expected to sharpen them.

13 Methods (variant)

As in the four-stimulus reproduction but with a $3 \times 3 = 9$ -patch field (75×75 image, corner cues S_1/S_9). Front-end: shared SE-ResNet to $\hat{o}_i \in \mathbb{R}^{128}$. Multiplicative gating $X' = \gamma(H) \odot X + \beta(H)$ then self-attention over the nine tokens, $Z = X' + AV$. Memory: per-patch spatial xLSTM, $d_{\text{mem}}=128$. Training: distributional actor-critic on sparse reward + V-JEPA. Analyses use $n=300$ trials/point; psychometrics fit a 4-parameter logistic. Decoders are 10-way (location) or 2-way (occurrence), trained on natural states. SDT uses hit/false-alarm rates at $\Delta=18^\circ$.

Figure 1A — Task timeline (vda9, 3×3 grid): 7 steps; corner cue at t_1 , 9 Gabors from t_3 , change (50%) at t_5

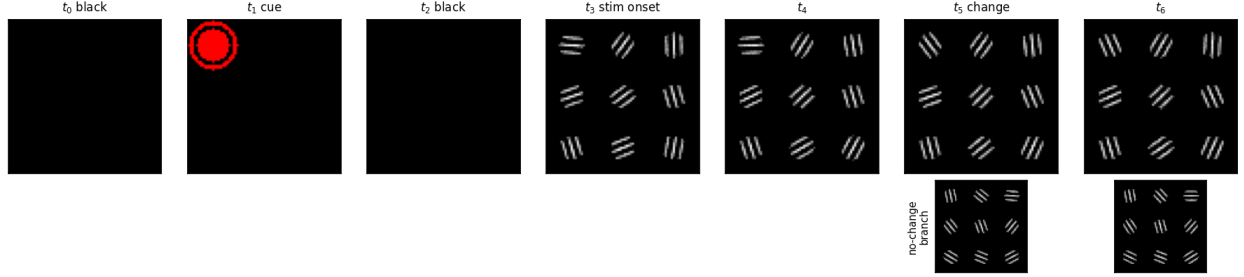


Figure 1B — Cue configs: validity ring × corner (S1/S9), empirical 3×3 change-location probability (our env)

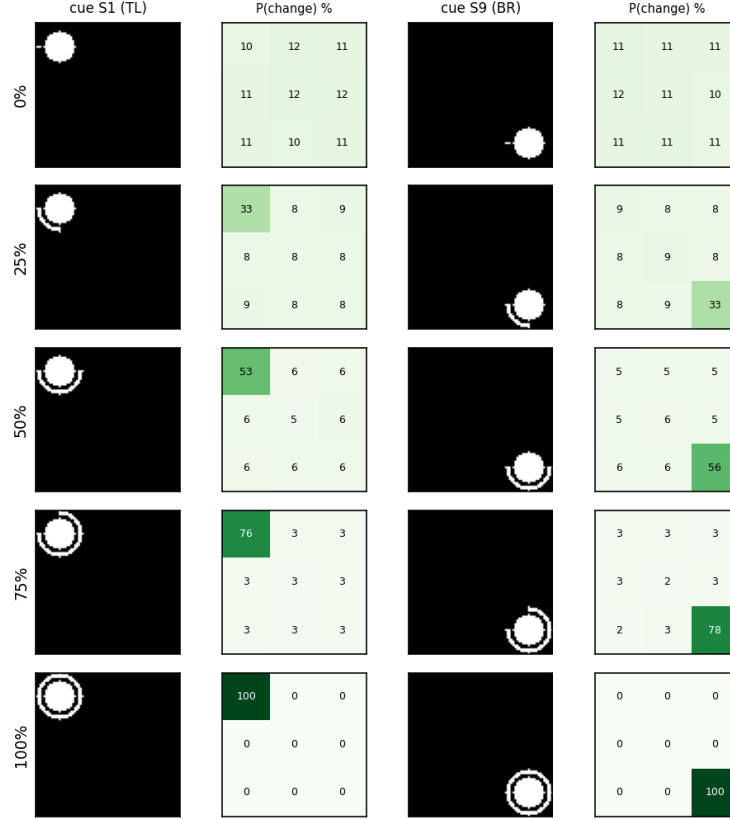


Figure 1: **Fig 1 — Task / environment (9-stim)**. Top: the 7-step timeline with change/no-change branch. Bottom: cue configurations (validity ring × corner) with the empirical 3 × 3 change-location probability our environment produces.

Figure 2 — Recurrent ViT schematic, 9-stimulus grid9 (affine_ew variant)

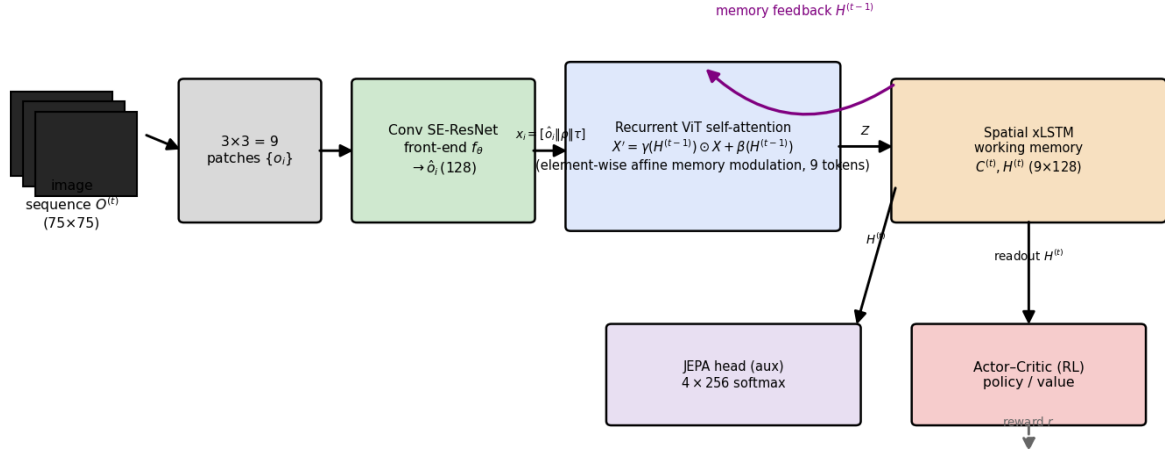


Figure 2: **Fig 2 — Model schematic (9-stim multiplicative-gating).** Nine-patch field $\rightarrow$ conv SE-ResNet $\rightarrow$ recurrent-ViT self-attention with memory gating ($X' = \gamma(H) \odot X + \beta(H)$) $\rightarrow$ spatial xLSTM $\rightarrow$ actor-critic (RL) and JEPA head.

Fig 3 (grid9, 9-stim) — affine_ew (iter 2999); validities incl 0-proportion, n=300/point

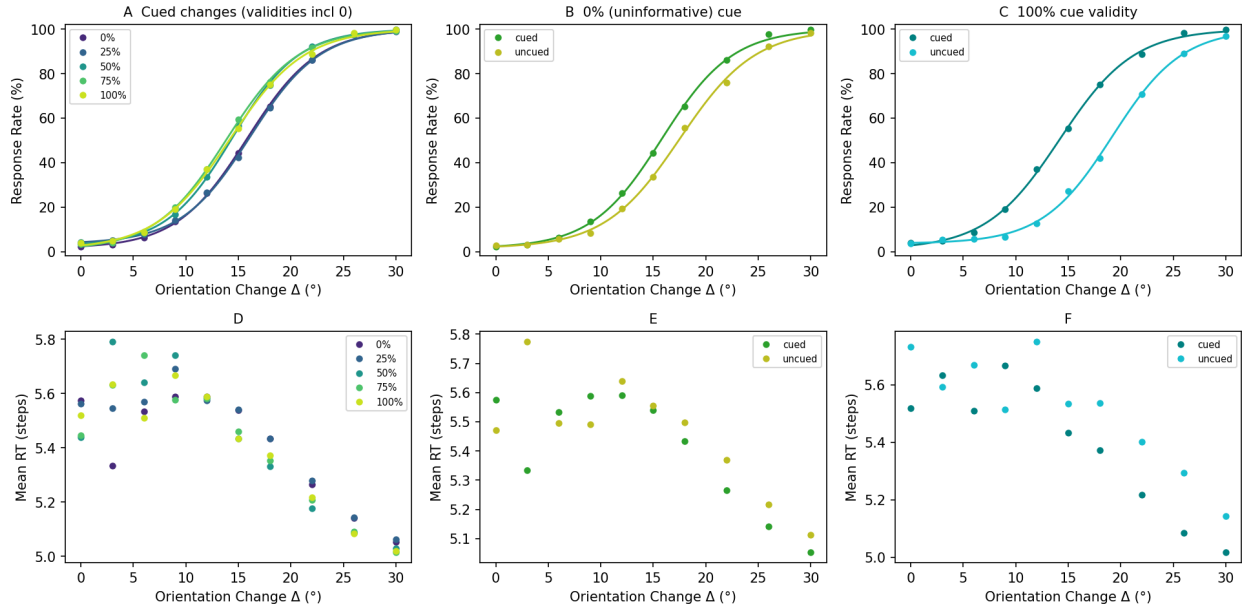


Figure 3: **Fig 3 — psychometrics and chronometrics.** A: cued response vs. Δ at five validities (incl. 0%). B: 0% cued vs. uncued. C: 100% cued vs. uncued (benefit larger). D–F: reaction time. Points $n=300$; lines are 4-parameter logistic fits.

Fig 3 fitted params — grid9 affine_ew (iter 2999); Position decreasing $\Rightarrow$ uses validity

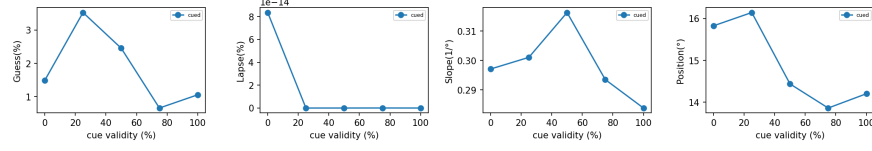


Figure 4: **Fitted psychometric parameters (cued) vs. validity.**

Fig 4A (grid9) — affine_ew: attention 3x3 heatmaps per timestep $\times$ validity (no-change cue-S1)

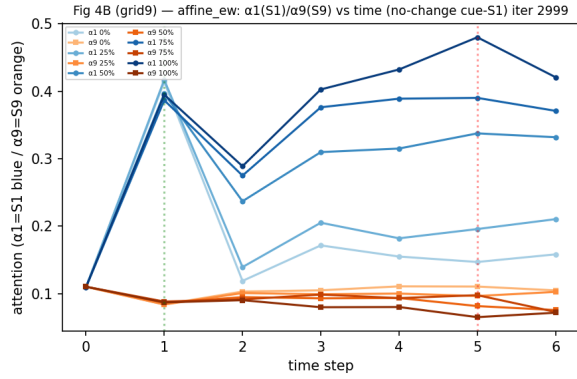
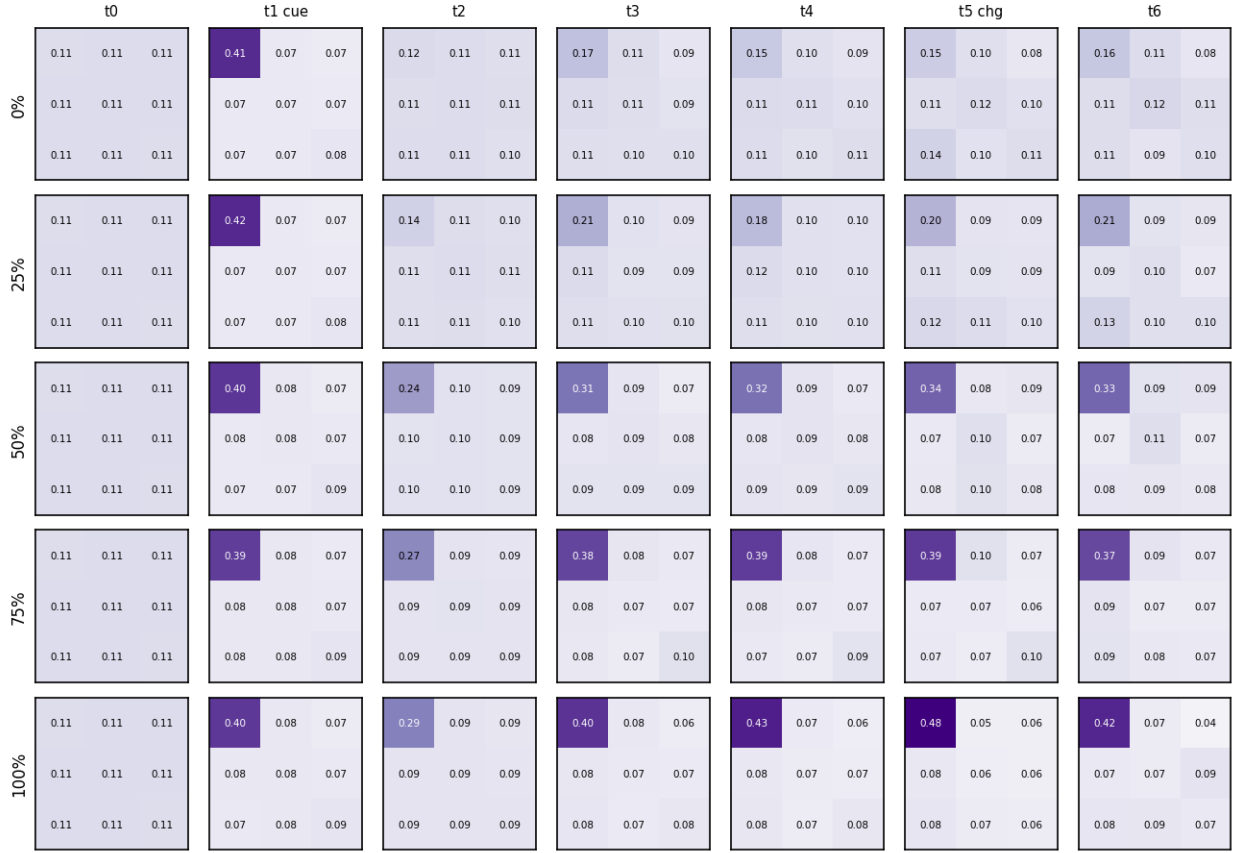


Fig 4B (grid9) — affine_ew: $\alpha_1(S1)/\alpha_9(S9)$ vs time (no-change cue-S1) iter 2999

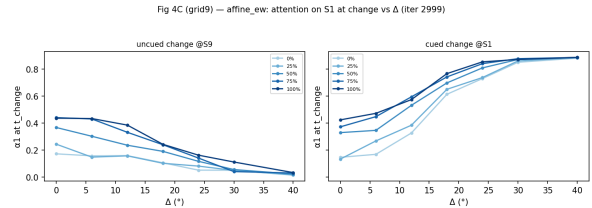


Fig 4C (grid9) — affine_ew: attention on S1 at change vs Δ (iter 2999)

Figure 5: **Fig 4 — intrinsic attention dynamics (9-stim).** Top: per-timestep 3×3 attention heatmaps by validity (no-change cue- S_1). Bottom left: α_1/α_9 vs. time. Bottom right: α_1 at change vs. Δ , uncued/cued.

Figs 5-7 analog --- feedback mechanism circuit & behaviour (affine_ew, iter 2999; n=250)

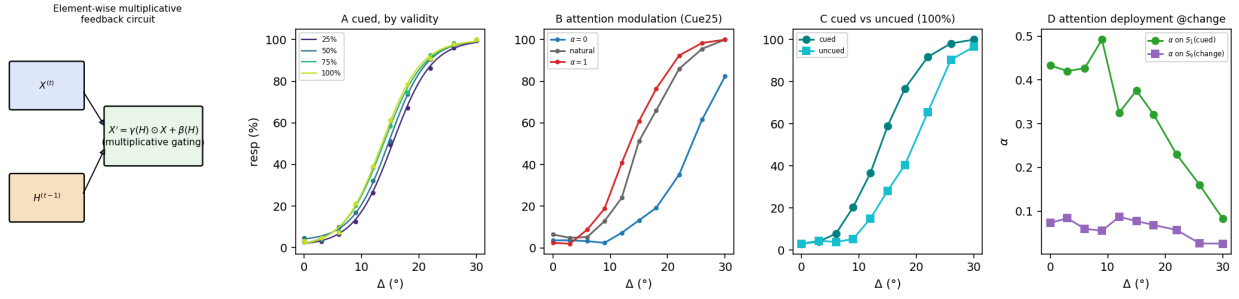


Figure 6: **Figs 5–7 analog — feedback circuit & behaviour (9-stim)**. Left: multiplicative-gating circuit. A: cued response by validity. B: enhance/suppress attention on the cued change. C: cued vs. uncued at 100%. D: attention on S_1 (cued) vs. S_9 (change) at the change frame.

Fig 5 A-F (grid9) — affine_ew: attention clamps at t_{change} (iter 2999, n=200)

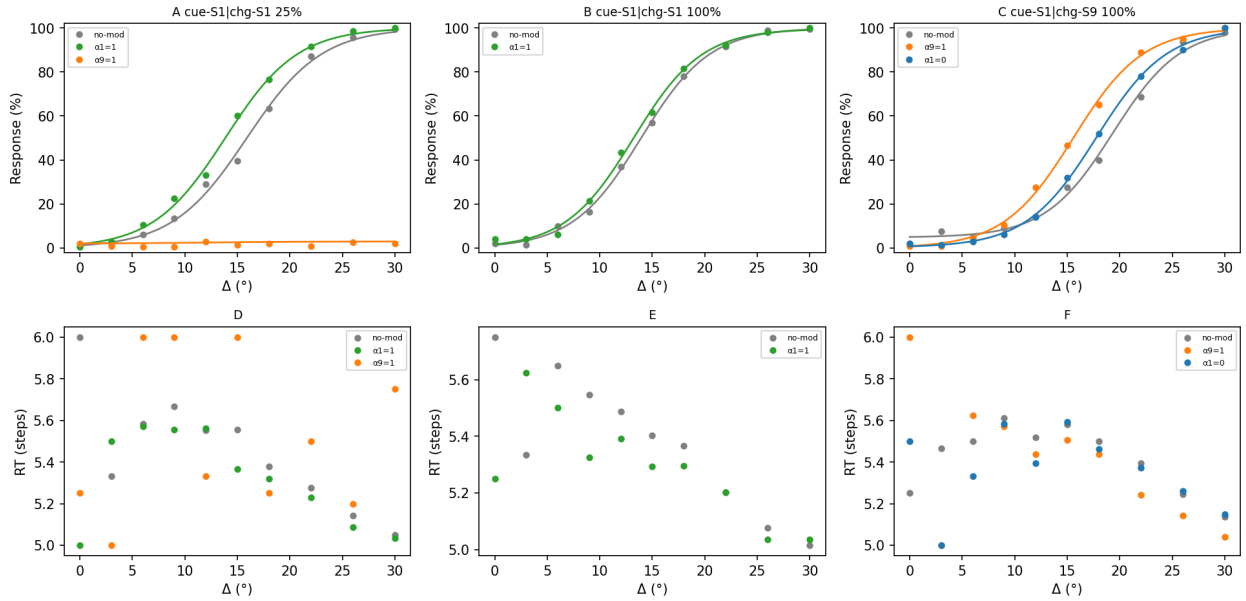


Figure 7: **Attention clamps at t_{change} (9-stim)**. Response (A–C) and RT (D–F) for clamps on S_1/S_9 .

Fig 5 G-K (grid9) — affine_ew: parametric attention manipulation → psychometric params (iter 2999)

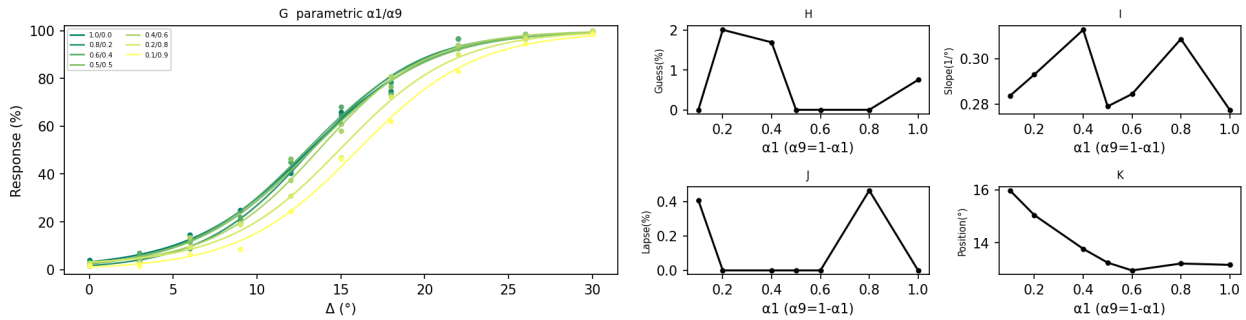


Figure 8: **Parametric manipulation (9-stim)**. Shifting attention $S_1 \leftrightarrow S_9$ grades the psychometric function; fitted parameters vs. α_1 .

Fig 8 --- decode change LOCATION from H, attention on S1 (affine_ew, iter 2999; test n=2000)

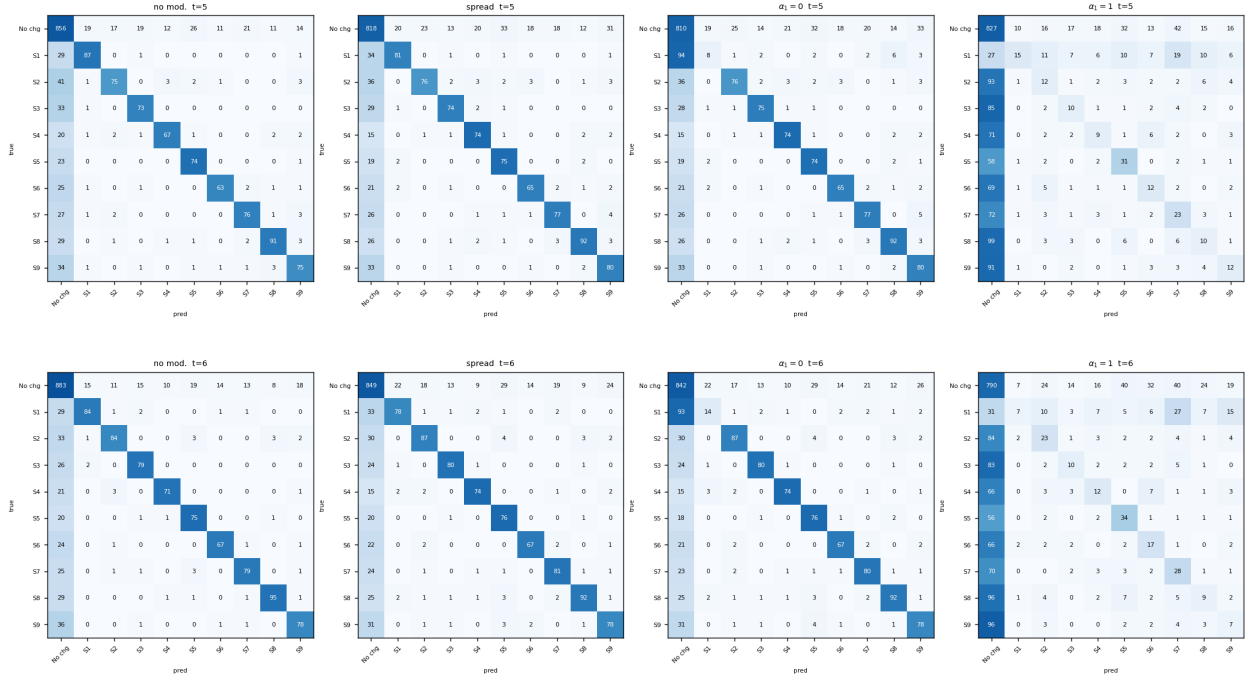


Figure 9: **Fig 8** — decode change LOCATION (10-way) from H , $t=5/6$, under $\{\text{no-mod, spread, } \alpha_1=0, \alpha_1=1\}$.

Fig 9 --- decode change OCCURRENCE from single patch h_1 (affine_ew, iter 2999; n=2000)

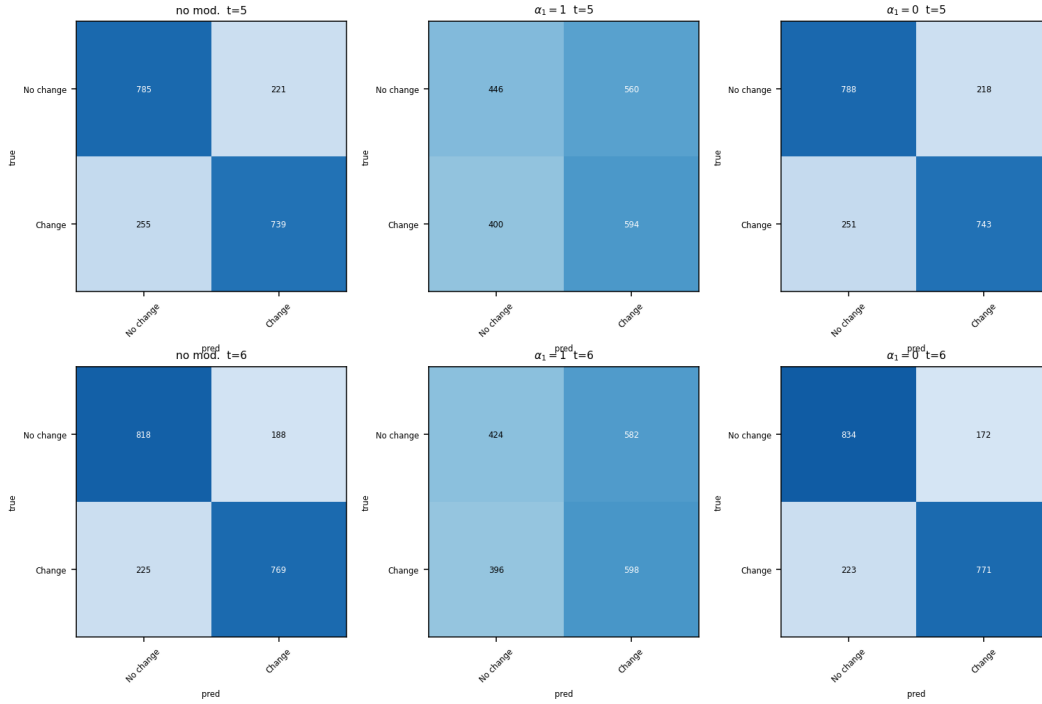


Figure 10: **Fig 9** — decode change OCCURRENCE from single patch h_1 , $t=5/6$.

Fig 10 --- decode from actor first activation layer (affine_ew, iter 2999; n=2000)

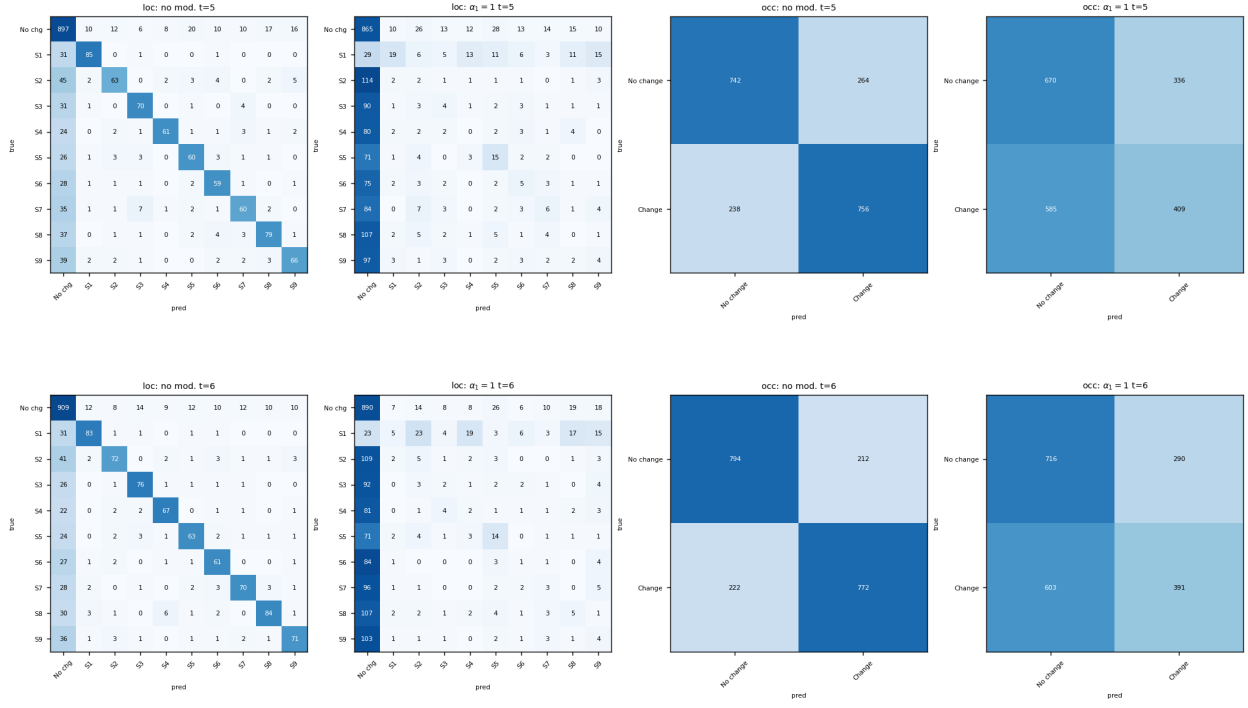


Figure 11: **Fig 10** — decode from actor first activation (location/occurrence).

Fig 11 --- decode S_1 -change vs no-change from actor activation, graded α_1 (affine_ew, iter 2999; n=2000)

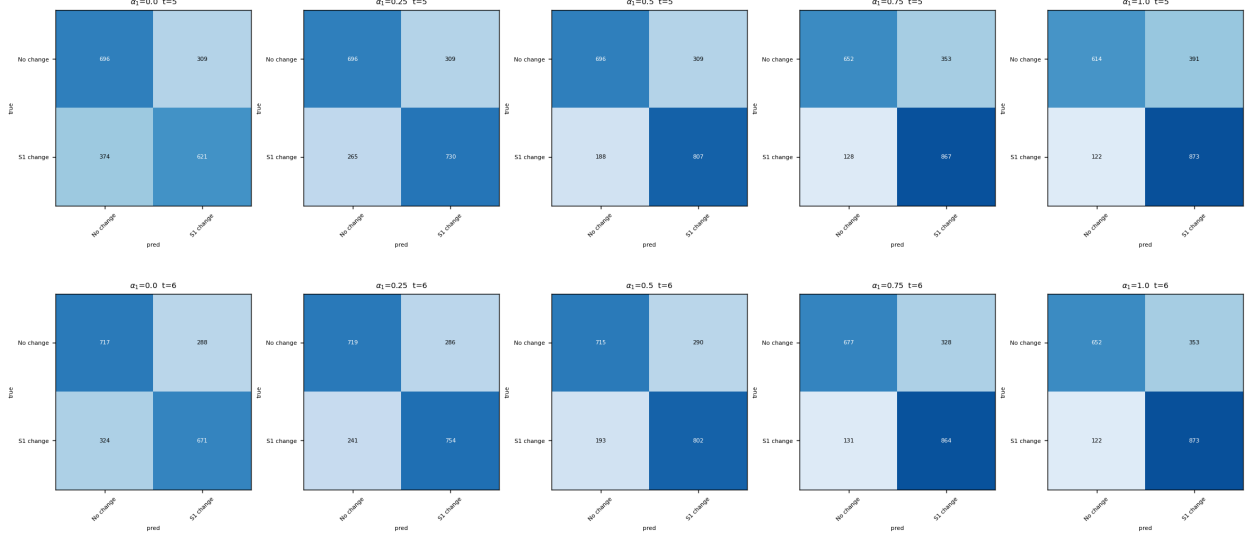


Figure 12: **Fig 11** — decode S_1 -change vs. no-change from actor activation, graded α_1 .

Fig 12 --- actor logit geometry at change frame (affine_ew, iter 2999; n=1500)

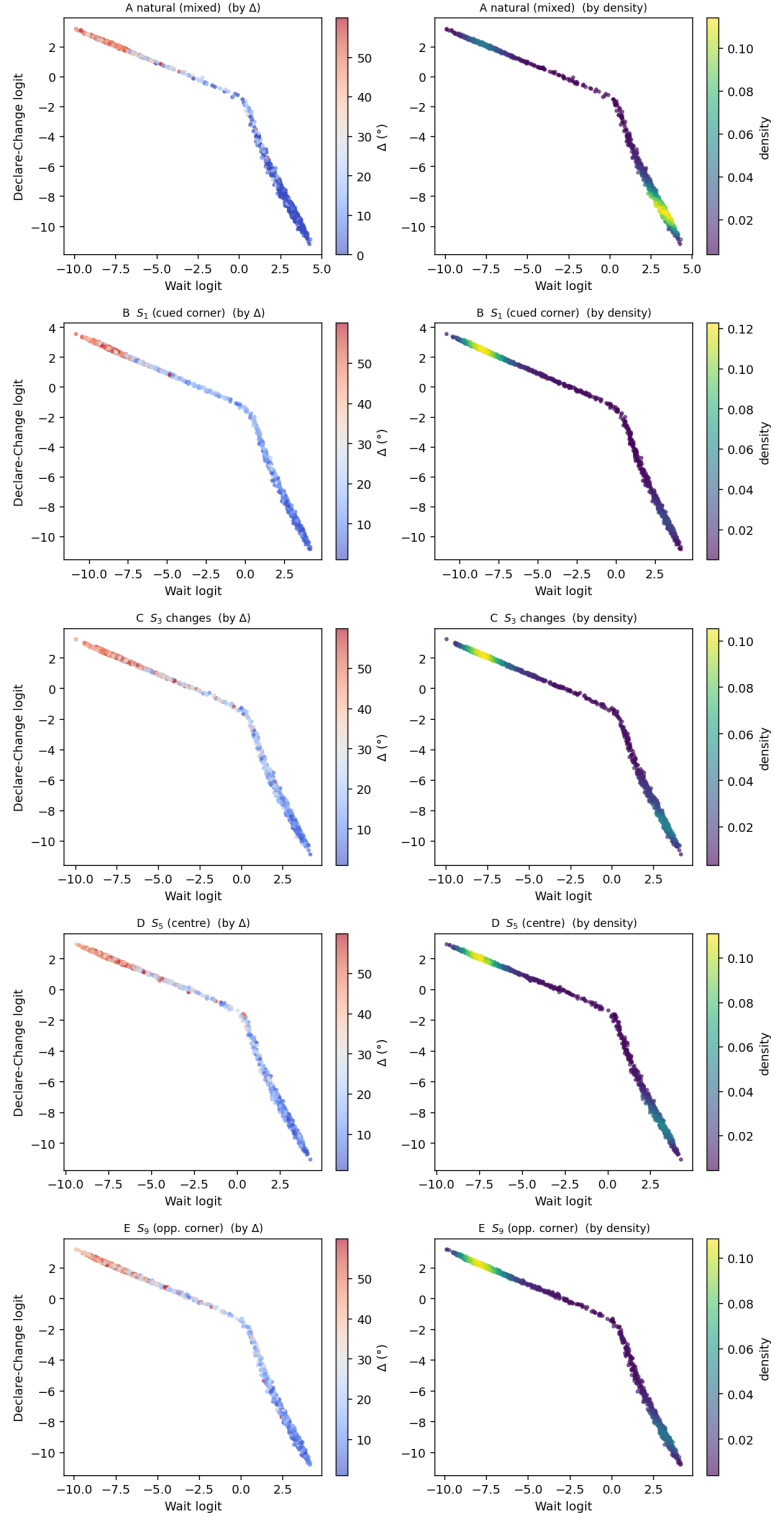


Figure 13: **Fig 12 — actor logit geometry (9-stim).** A: natural. B-E: changes at $S_1/S_3/ S_5/S_9$. Coloured by Δ (left) and density (right).

Fig 13 --- actor logits under S_1 attention modulation (affine_ew, iter 2999; $n=1500$)

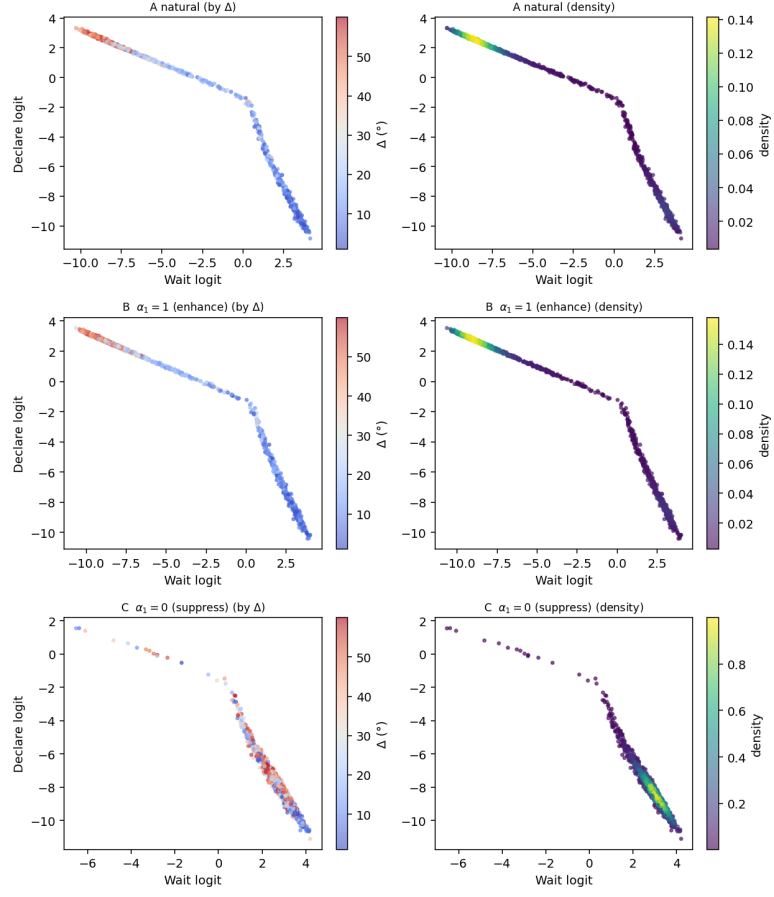


Figure 14: **Fig 13** — actor logits under S_1 attention modulation.

Fig 14 --- TD error & value under change magnitude and attention (affine_ew, iter 2999)

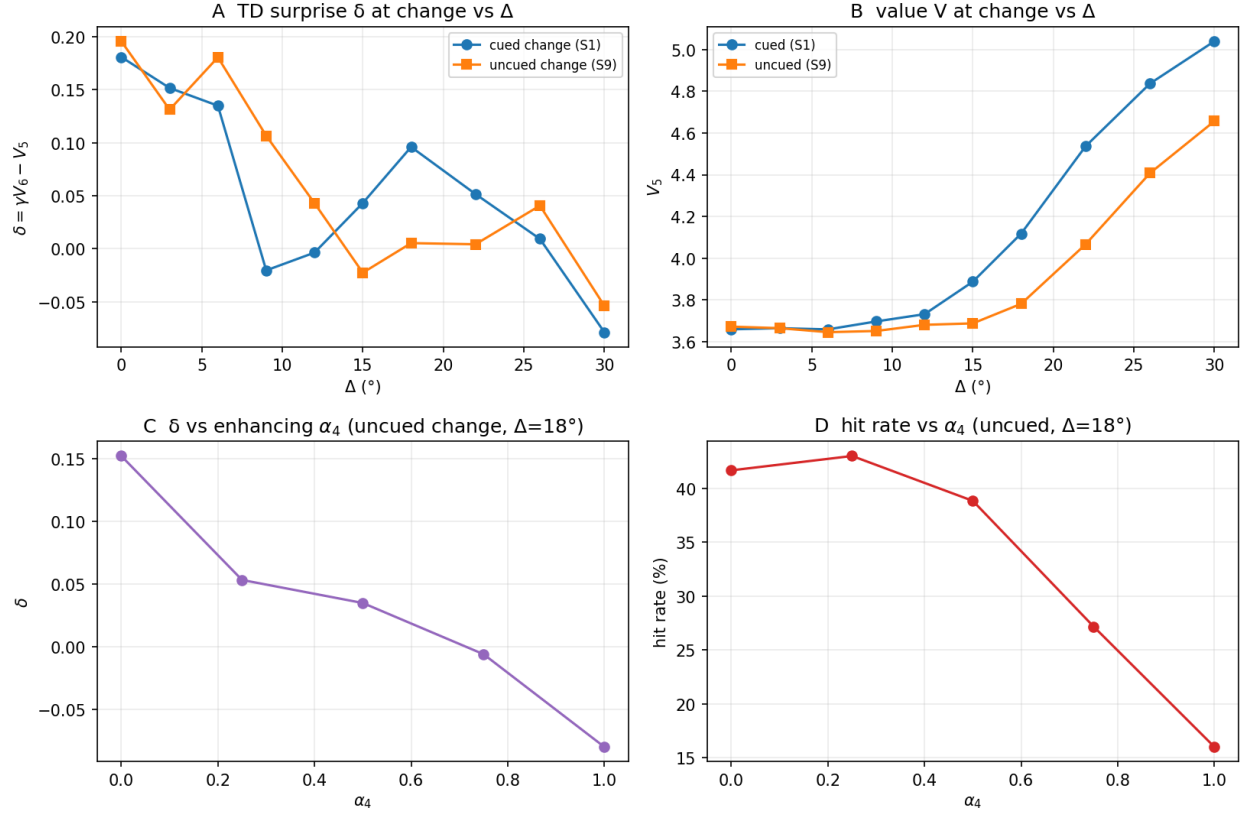


Figure 15: Fig 14 — TD error & value (9-stim).

Fig 15 --- entropy & criterion-value coupling under attention (affine_ew, iter 2999; $\Delta=18^\circ$, $n=400$)

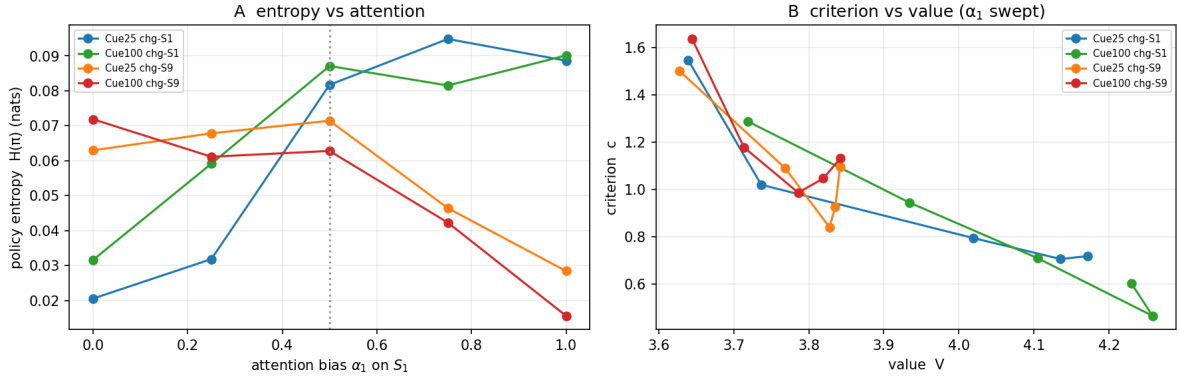


Figure 16: Fig 15 — entropy & criterion-value coupling (9-stim).

Fig 16 --- SDT criterion & sensitivity vs attention bias on S_1 (affine_ew, iter 2999; $\Delta=18^\circ$, $n=300$)

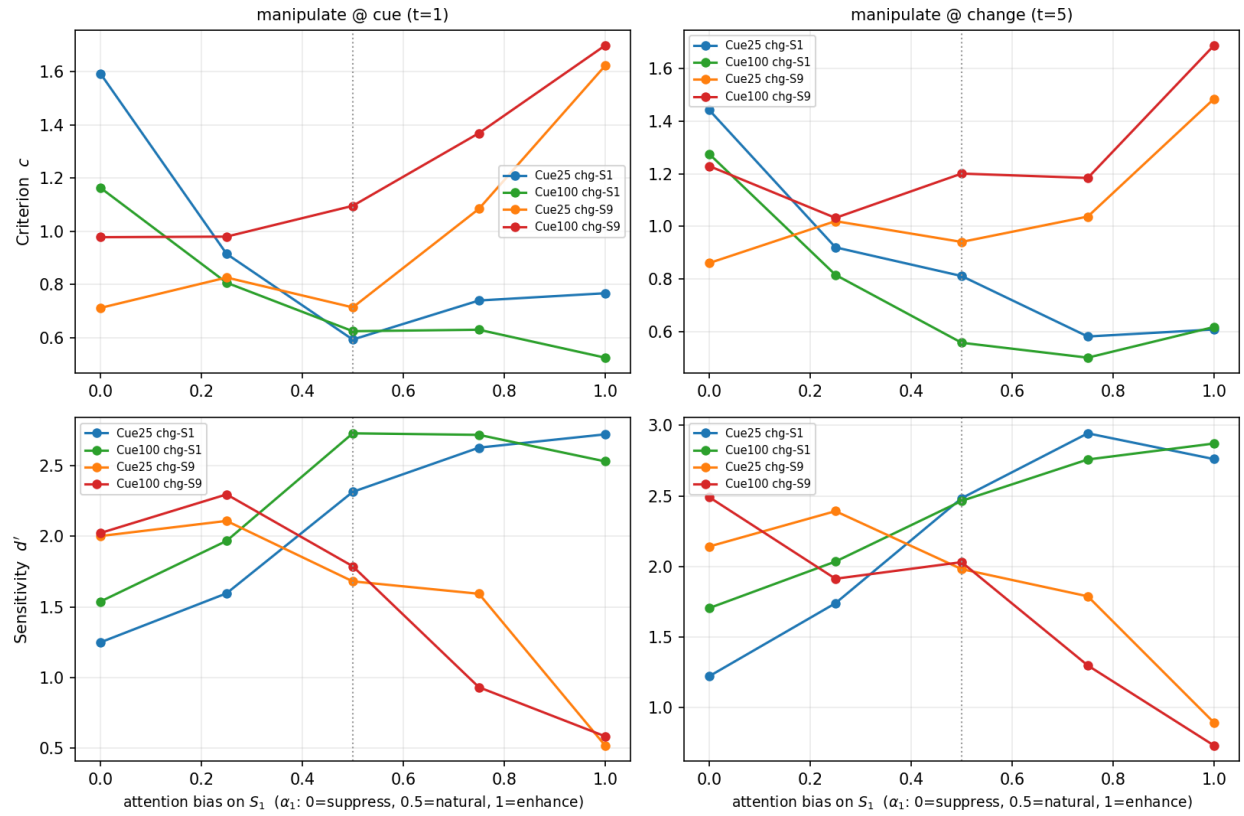


Fig 17 analog --- supervised vs RL (RL column reproduced; supervised = out-of-scope baseline) (affine_ew, iter 2999)

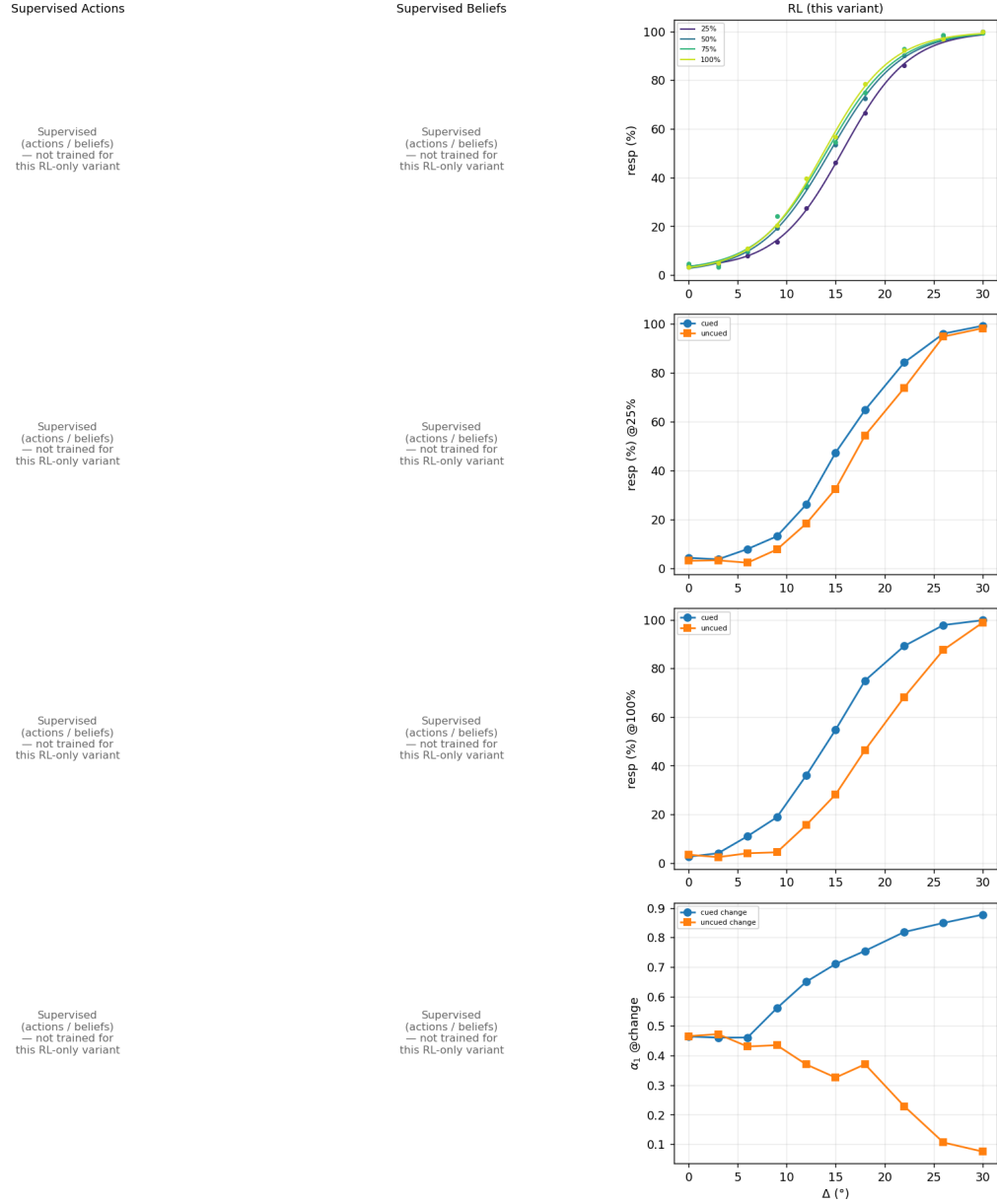


Figure 18: Fig 17 analog — supervised vs. RL (9-stim).